

Distributed List Coloring from Lists of Size $d(d + 1)$

AI-generated research note

30 September 2026

Abstract

Let a finite simple graph be supplied with an orientation of maximum outdegree at most $d \geq 2$. We give a uniform deterministic distributed algorithm that properly colors the graph from arbitrary vertex lists of size at least $d(d + 1)$ in $O(\log^* M + d^2)$ rounds, where distinct vertex identifiers lie in a commonly known namespace of size M . The algorithm uses only information obtained by reading outgoing neighbors. Indegree is unrestricted, and the round bound has no dependence on the universe of list colors. The construction removes one color from the sufficient list size $d(d + 1) + 1$ obtained by an earlier two-sweep method. Its additional ingredient is a canonical-prefix argument: the only possible equality obstruction lies in a fixed independent set, and each vertex of that set has two list colors absent from the entire input lists of all but one outgoing neighbor. We give all combinatorial, preprocessing, and round-complexity proofs.

1 Context and contribution

List coloring assigns each vertex v a color from its own prescribed list, with different colors at adjacent vertices. The bound in this note is a bound on the size of *each list*, not on the total number of distinct colors appearing anywhere in the input or output. In particular, different vertices may have disjoint lists.

The starting point is the two-sweep approach to distributed coloring. Barenboim and Elkin introduced increasing and decreasing color-class sweeps for defective coloring [1]. Fuchs and Kuhn extended this approach to list defective coloring [3, 4]: a vertex first chooses a set of candidate colors, and a reverse sweep selects its final color. We use this candidate-set architecture and the restriction to outgoing neighbors from their oriented setting. The preliminary reduction of identifier colors to a quadratic palette uses Linial’s separating-set method [5], including a finite-field polynomial construction of Erdős, Frankl, and Füredi [2]. A complete construction and analysis appear in Section 4.

For a precise comparison, the case with all allowed defects zero and auxiliary parameter $\varepsilon = 0$ in Fuchs and Kuhn’s theorem [3, Theorem 1.1] (also [4, Theorem 2.1]) permits lists of a common size s when

$$s > \max\{p, s/p\} d$$

for an integer parameter $p \geq 1$. Taking $p = d + 1$ and $s = d(d + 1) + 1$ satisfies this condition, since $\max\{p, s/p\} = d + 1$. Their earlier 2024 work therefore already provides this comparison bound.

We show that $s = d(d + 1)$ suffices for $d \geq 2$. The proof identifies the exact equality case in a candidate-set counting argument, reserves two colors at every vertex in a statically defined exceptional set, and runs the two sweeps on the remaining vertices. Crucially, the reserved colors are isolated from the other neighbors’ *full input lists*; isolation from

their initial candidate sets would not suffice. This is a sufficient bound, with no claim of optimality.

2 Model and main theorem

The empty graph is trivially colored, so assume $n = |V| \geq 1$. The graph $G = (V, E_G)$ is finite and simple, with exactly one supplied direction on each edge. Write $v \rightarrow u$ for a directed edge and set

$$N^+(v) = \{u : v \rightarrow u\}, \quad \deg^+(v) = |N^+(v)|.$$

A common integer $d \geq 2$ bounds every outdegree. There is no restriction on indegree, connectedness, or directed cycles. The outgoing edges at a vertex have distinct local port labels, ordered locally; no consistency between port orders at different vertices is required.

Each vertex v has a distinct identifier in $[M] = \{1, \dots, M\}$, where $M \geq 1$ is a common known bound. It also receives a finite set $L_{\text{in}}(v)$ of distinct color names. Names have finite representations and a common total order, with consistent equality and comparison available as local computation. For example, names may be arbitrary nonnegative integers. There is no supplied bound on their magnitude, on the union of the lists, or on the lengths of their representations.

We use the synchronous **LOCAL** convention of unrestricted local computation and message size; see Linial [5] for the distributed locality framework. More specifically, the algorithm obeys the following *forward-information restriction*. Initially, a vertex knows its own identifier, input list, outgoing ports, and the common parameters. In one round it can read the previous-round state of every outgoing neighbor and then update its own state. Its transition reads no incoming-neighbor state or incoming incidence information. Thus *forward* describes which vertices may be observed: for $v \rightarrow u$, v reads information from u . It is not a requirement that messages travel from tail to head. Ordinary two-way communication on the underlying undirected graph can implement these reads.

All finite computations between rounds are free in this model. In particular, finite searches used below are not claimed to be computationally efficient. Uniformity means that one finite program, given d and M , performs the stated constructions and terminates on every legal input; no family of graph-dependent advice is supplied. All vertices use common phase lengths determined by these parameters.

All logarithms without a subscript are base two; $\ln$ denotes the natural logarithm. For $x \geq 1$, $\log^* x$ is the least number of iterations of the base-two logarithm needed to reach a value at most 1.

Theorem 2.1. *Under the preceding conventions, suppose that*

$$|L_{\text{in}}(v)| \geq d(d+1) \quad \text{for every } v \in V.$$

There is a uniform deterministic forward-information algorithm that computes a coloring φ satisfying

$$\varphi(v) \in L_{\text{in}}(v) \quad (v \in V), \quad \varphi(v) \neq \varphi(u) \quad (v \rightarrow u)$$

in $O(\log^ M + d^2)$ rounds, with absolute constants. The round bound is independent of the color-name universe. If $n = |V|$ and $M \leq \max\{2, n\}^C$ for a fixed constant C , the bound is $O(\log^* n + d^2)$.*

The restriction $d \geq 2$ is necessary for this formula: a cyclically oriented triangle has outdegree one and has no proper coloring from the common two-element list.

We first prove the list-coloring stage from a proper ordered coloring, then construct that ordered coloring from the identifiers. A *proper ordered coloring* is a map $\lambda : V \rightarrow [K]$ such that $\lambda(v) \neq \lambda(u)$ on every edge. The order on $[K]$ schedules vertices; it imposes no monotonicity on the supplied edge directions.

Proposition 2.2. *Suppose a proper ordered coloring $\lambda : V \rightarrow [K]$ and the common bound $K \geq 1$ are already available. Under the list-size and outdegree assumptions of Theorem 2.1, a deterministic forward-information algorithm computes a proper list coloring in at most $2K + 4$ additional rounds.*

3 The list-coloring stage

3.1 Canonical prefixes and exceptional vertices

Fix $d \geq 2$ and put $k = d(d + 1)$. For a finite ordered set S and an integer r with $0 \leq r \leq |S|$, let $\text{first}_r(S)$ denote its r least elements. Retain

$$L(v) = \text{first}_k(L_{\text{in}}(v)), \quad |L(v)| = k.$$

Every set and count in the setup below is defined from these lists, the orientation, and λ , before either sweep starts.

Let

$$\mathcal{F} = \{f : |N^+(f)| = d \text{ and } \lambda(u) > \lambda(f) \text{ for every } u \in N^+(f)\}.$$

For $f \in \mathcal{F}$, define its canonical prefix

$$B(f) = \text{first}_{d+1}(L(f)) = \text{first}_{d+1}(L_{\text{in}}(f)). \quad (3.1)$$

The second equality holds because truncation retained the least $k \geq d + 1$ elements.

A vertex e is *exceptional* if it has exactly d outgoing neighbors, all of them have smaller rank than e , all of them belong to $\mathcal{F}$, and

$$L(e) = \bigsqcup_{f \in N^+(e)} B(f). \quad (3.2)$$

Here the disjoint-union notation requires the d prefixes to be pairwise disjoint. Write $\mathcal{E}$ for the exceptional set and $\mathcal{N} = V \setminus \mathcal{E}$ for the set of normal vertices.

Lemma 3.1. *The sets $\mathcal{E}$ and $\mathcal{F}$ are disjoint, every outgoing neighbor of an exceptional vertex is normal, and $\mathcal{E}$ is an independent set in the underlying undirected graph.*

Proof. A vertex of $\mathcal{F}$ has d outgoing neighbors of larger rank, whereas a vertex of $\mathcal{E}$ has d outgoing neighbors of smaller rank. Since $d > 0$, it cannot have both properties. Every outgoing neighbor of a vertex of $\mathcal{E}$ belongs to $\mathcal{F}$, hence does not belong to $\mathcal{E}$. Therefore no directed edge has both endpoints in $\mathcal{E}$; every underlying edge has a direction, so the set is independent. $\square$

A vertex in $\mathcal{F}$ may have outgoing neighbors in $\mathcal{E}$. In that event, deletions made below may change its eventual candidate set. The proof will identify that candidate set with $B(f)$ only under an additional equality condition. Membership in $\mathcal{E}$ does not assert that a later sweep would actually fail there.

3.2 Two reserved colors insulated from full input lists

Lemma 3.2 (Prefix isolation). *For every $e \in \mathcal{E}$, one can deterministically choose a distinguished outgoing neighbor $f(e)$ and a two-element set $P(e)$ such that*

$$\begin{aligned} P(e) &\subseteq L(e) \cap B(f(e)), \\ P(e) \cap L_{\text{in}}(g) &= \emptyset \quad \text{for every } g \in N^+(e) \setminus \{f(e)\}. \end{aligned} \tag{3.3}$$

The choice uses only e 's list, its outgoing prefixes, and its local port order.

Proof. Let $Q(e) = \text{first}_{d+1}(L(e))$. The d disjoint prefixes in (3.2) partition $L(e)$. Since $Q(e)$ has $d+1$ elements, at least one of these prefixes contains at least two elements of $Q(e)$. Among these eligible neighbors, let $f(e)$ be the one on the least local outgoing port. Let $P(e)$ consist of the two least elements of $Q(e) \cap B(f(e))$. This proves the first inclusion.

Fix any other outgoing neighbor g , and put $q = \max Q(e)$. Since q is the $(d+1)$ st least element of $L(e)$, every $(d+1)$ -element subset of $L(e)$ has maximum at least q . In particular,

$$q \leq \max B(g).$$

For $x \in P(e)$, we have $x \leq q$, and disjointness of the prefixes gives $x \notin B(g)$. On the other hand, (3.1) implies that every element of $L_{\text{in}}(g) \setminus B(g)$ is strictly larger than $\max B(g)$. Thus x belongs neither to $B(g)$ nor to its complement within $L_{\text{in}}(g)$, proving $x \notin L_{\text{in}}(g)$. $\square$

Consequently, once the outgoing neighbors of e have chosen arbitrary colors from their own input lists, all but $f(e)$ avoid both colors in $P(e)$. The single color of $f(e)$ can rule out at most one of them. This guarantee survives any changes to those neighbors' candidate sets. Computing the pair requires reading only the canonical prefixes; the full-list exclusion is a proved property of the unread list tails.

3.3 Deleting reserved pairs and choosing candidate sets

For a normal vertex v , partition its outgoing neighbors into exceptional neighbors, lower-ranked normal neighbors, and higher-ranked normal neighbors. Let their respective numbers be

$$\begin{aligned} t(v) &= |N^+(v) \cap \mathcal{E}|, \\ a(v) &= |\{u \in N^+(v) \cap \mathcal{N} : \lambda(u) < \lambda(v)\}|, \\ b(v) &= |\{u \in N^+(v) \cap \mathcal{N} : \lambda(u) > \lambda(v)\}|. \end{aligned} \tag{3.4}$$

Properness of λ implies that these are all the possibilities, and hence

$$a(v) + b(v) + t(v) = \deg^+(v) \leq d. \tag{3.5}$$

Delete the reserved pairs of outgoing exceptional neighbors:

$$L'(v) = L(v) \setminus \bigcup_{e \in N^+(v) \cap \mathcal{E}} P(e). \tag{3.6}$$

Overlaps among the deleted pairs can only help, so

$$|L'(v)| \geq d(d+1) - 2t(v). \tag{3.7}$$

The increasing sweep processes rank classes $1, 2, \dots, K$. Vertices in the same class act simultaneously. A normal vertex v chooses the candidate set, or *advertisement*,

$$A(v) = \text{first}_{b(v)+1} \left(L'(v) \setminus \bigcup_{\substack{u \in N^+(v) \cap \mathcal{N} \\ \lambda(u) < \lambda(v)}} A(u) \right). \quad (3.8)$$

Only advertisements already constructed at smaller ranks occur on the right. Exceptional vertices do nothing during this sweep.

Lemma 3.3. *Every choice in (3.8) is possible. In particular, $|A(v)| = b(v) + 1 \leq d + 1$ for every normal vertex.*

Proof. Induct on the processed rank classes. Suppose all required lower-ranked advertisements have been constructed with the stated sizes. Fix a vertex v in the current class, and abbreviate $a = a(v)$, $b = b(v)$, and $t = t(v)$. By (3.7) and the induction hypothesis, the number of available elements in (3.8) is at least

$$d(d+1) - 2t - a(d+1). \quad (3.9)$$

First suppose $t \geq 1$. Using $a \leq d - b - t$, the difference between (3.9) and the required $b + 1$ is at least

$$\begin{aligned} d(d+1) - 2t - (d - b - t)(d+1) - (b+1) &= db + (d-1)t - 1 \\ &\geq 0, \end{aligned}$$

where the last inequality uses $d \geq 2$. This includes the tight case $d = 2$, $t = 1$, $a = 1$, $b = 0$.

Next suppose $t = 0$ and $a \leq d - 1$. Put $s = d - a \geq 1$. The lower bound (3.9) is $s(d+1)$. Since $b \leq s$ and $s(d+1) \geq s+1$, there are at least $b+1$ available elements.

The only remaining case is $t = 0$ and $a = d$. Equation (3.5) forces $b = 0$, so only one element is needed. Suppose for a contradiction that none is available. Since $t = 0$, we have $L'(v) = L(v)$. The d lower outgoing advertisements then cover $L(v)$, giving

$$d(d+1) = |L(v)| \leq \left| \bigcup_{u \in N^+(v)} A(u) \right| \leq \sum_{u \in N^+(v)} |A(u)| \leq d(d+1). \quad (3.10)$$

Every inequality is an equality. The union contains $L(v)$ and has the same cardinality, so it equals $L(v)$. The sum of advertisement sizes is $d(d+1)$, so every one of the d advertisements has size $d+1$. Equality between union size and the sum of sizes says that they are pairwise disjoint. Thus they partition $L(v)$.

For each such neighbor u , the already proved identity $|A(u)| = b(u) + 1 = d + 1$ implies $b(u) = d$. Applying (3.5) at u gives $a(u) = t(u) = 0$. Consequently u has exactly d outgoing neighbors, all normal and of larger rank; in particular $u \in \mathcal{F}$. No pair was deleted from $L(u)$, and no smaller-ranked advertisement was excluded at u . The stipulated least-element rule in (3.8) therefore gives

$$A(u) = \text{first}_{d+1}(L(u)) = B(u).$$

We have proved that v has d smaller-ranked outgoing neighbors in $\mathcal{F}$ whose canonical prefixes partition $L(v)$. These are precisely the conditions defining $\mathcal{E}$. This contradicts $v \in \mathcal{N}$. The putative failure is impossible, and the induction is complete. $\square$

The use of *higher normal* neighbors in $b(v)$, and the canonical least-element choice in the increasing sweep, are both needed in the equality argument. There is no assertion that an arbitrary vertex of $\mathcal{F}$ keeps its canonical prefix.

3.4 The decreasing sweep and exceptional completion

Process normal vertices in rank order $K, K-1, \dots, 1$. At v , choose the least color in

$$A(v) \setminus \{\varphi(u) : u \in N^+(v) \cap \mathcal{N}, \lambda(u) > \lambda(v)\}. \quad (3.11)$$

There are $b(v)$ higher-ranked normal outgoing neighbors and $|A(v)| = b(v) + 1$, so the choice is possible.

After this sweep, all normal vertices have final colors. For each exceptional vertex e , simultaneously set

$$\varphi(e) = \min(P(e) \setminus \{\varphi(f(e))\}). \quad (3.12)$$

Lemma 3.1 guarantees that $f(e)$ is normal, so its color is already defined. A single color cannot exhaust the two-element set $P(e)$.

Lemma 3.4. *The resulting φ is a proper coloring and $\varphi(v) \in L_{\text{in}}(v)$ at every vertex.*

Proof. For a normal vertex, the construction gives

$$\varphi(v) \in A(v) \subseteq L'(v) \subseteq L(v) \subseteq L_{\text{in}}(v).$$

For an exceptional vertex, it gives $\varphi(e) \in P(e) \subseteq L(e) \subseteq L_{\text{in}}(e)$. It remains to verify every directed edge $v \rightarrow u$.

If $v, u \in \mathcal{N}$ and $\lambda(u) < \lambda(v)$, the increasing sweep chose $A(v)$ disjoint from $A(u)$, so their final colors differ. If $v, u \in \mathcal{N}$ and $\lambda(u) > \lambda(v)$, the decreasing choice at v explicitly excluded $\varphi(u)$. There is no equal-rank case.

If $v \in \mathcal{N}$ and $u \in \mathcal{E}$, then $A(v) \subseteq L'(v)$ is disjoint from $P(u)$ by (3.6). Since $\varphi(u) \in P(u)$, the edge is proper.

If $v \in \mathcal{E}$ and $u \in \mathcal{N}$, the choice in (3.12) excludes $\varphi(u)$ when $u = f(v)$. Otherwise Lemma 3.2 gives $P(v) \cap L_{\text{in}}(u) = \emptyset$, whereas $\varphi(u) \in L_{\text{in}}(u)$ and $\varphi(v) \in P(v)$. Again the colors differ.

There are no edges with both endpoints exceptional, by Lemma 3.1. These cases cover every underlying edge. $\square$

3.5 Communication schedule

Here is a complete schedule under the read-previous-state convention of Section 2. Vertices retain any state needed in later rounds.

1. In the first setup round, each vertex reads its outgoing neighbors' ranks and determines membership in $\mathcal{F}$. If it belongs to $\mathcal{F}$, it computes its canonical prefix locally.
2. In the second round, it reads its outgoing neighbors' membership in $\mathcal{F}$, together with the prefixes of those in $\mathcal{F}$. It can now test membership in $\mathcal{E}$; if exceptional, it chooses $f(e)$ and $P(e)$.
3. In the third round, it reads its outgoing neighbors' exceptional flags, together with the pairs of those in $\mathcal{E}$. Every normal vertex can now determine $a(v), b(v), t(v)$ and $L'(v)$.
4. Allocate K rounds to the increasing sweep, one per rank class. At its scheduled round a vertex reads the stored advertisements of its smaller-ranked outgoing normal neighbors.
5. Allocate K rounds to the decreasing sweep. At its scheduled round a vertex reads the final colors of its higher-ranked outgoing normal neighbors.

6. One final round gives each exceptional vertex its distinguished outgoing neighbor's final color, allowing (3.12).

The total is $3 + K + K + 1 = 2K + 4$ rounds. Adjacent vertices have different ranks, and every dependency in each sweep points to a rank class already processed in that sweep. This remains true on directed cycles. No distinguished neighbor needs to learn which incoming vertices selected it. All communicated objects are finite and every operation is deterministic, using ordered lists or local port order to break ties. Together with Lemmas 3.3 and 3.4, this proves Proposition 2.2.

4 Self-contained construction of the ordered coloring

We now supply the preprocessing needed for the main theorem. The separating-set transformation is Linial's color-reduction method [5, Section 4]; the polynomial sets used in its last step come from Erdős, Frankl, and Füredi [2, Example 3.2]. We prove the required properties here, including a round bound uniform in d , and specify finite deterministic searches so that no nonuniform advice is required. These computations use identifiers and temporary labels, not the input list colors.

Proposition 4.1. *From identifiers in the common namespace $[M]$, one can compute a proper ordered coloring $\lambda : V \rightarrow [K]$ with*

$$K < 256(d + 1)^2$$

in at most $2 \log^ M + 3$ forward-information rounds.*

4.1 A separating family and one-round reduction

Put $u = d + 1$ and $A = 8u^2$. For any integer $h \geq 2$, let

$$m = A \lceil \log h \rceil.$$

We claim that there are sets $S_1, \dots, S_h \subseteq [m]$ such that

$$S_i \setminus \bigcup_{j \in J} S_j \neq \emptyset \quad \text{whenever } i \in [h], \ J \subseteq [h] \setminus \{i\}, \ |J| \leq d. \quad (4.1)$$

To prove existence, independently include each point in each set with probability $1/u$. For a fixed pair (i, J) , a given point belongs to S_i and to none of the sets indexed by J with probability at least

$$\frac{1}{u} \left(1 - \frac{1}{u}\right)^d = \frac{1}{u} \left(1 + \frac{1}{d}\right)^{-d} \geq \frac{1}{eu}.$$

The last inequality follows from $\ln(1 + x) \leq x$ for $x \geq 0$. Independence across points and $1 - z \leq e^{-z}$ imply that the probability that this pair has no witness is at most

$$\exp\left(-\frac{m}{eu}\right) \leq \exp(-2u \ln h) = h^{-2u}. \quad (4.2)$$

Indeed, $m/(eu) = (8/e)u \lceil \log h \rceil \geq 2u \ln h$, using $e < 3$ and $\log h \geq \ln h$.

There are at most $h^{d+1} = h^u$ pairs (i, J) . For fixed i , encode J by its increasing list of elements, followed by copies of i until the sequence has length d . This is an injection into $[h]^d$, since $i \notin J$. The union bound and (4.2) put the probability of any failed pair at most $h^{-u} < 1$. A family satisfying (4.1) therefore exists.

To select a common family deterministically, order all h -tuples of subsets of $[m]$ lexicographically by their incidence bit strings. Enumerate them until all the finitely many conditions in (4.1) hold. The existence proof guarantees termination, and the common parameters guarantee that every vertex finds the same family.

Suppose the current proper coloring has labels in $[h]$. A vertex of label i reads its outgoing neighbors' labels in one round and chooses the least element of

$$S_i \setminus \bigcup_{w \in N^+(v)} S_{\text{old}(w)}. \quad (4.3)$$

Neighbor labels differ from i ; repeated labels at different neighbors cause no problem because there are at most d distinct ones. The choice is possible by (4.1). For an edge $v \rightarrow w$, the new label of v is outside $S_{\text{old}(w)}$, whereas the new label of w belongs to that set. Hence the new coloring is proper and has palette $[m]$.

4.2 Bounding the number of reductions uniformly

Let

$$H_0 = 16A \lceil \log(4A) \rceil, \quad f(x) = A \lceil \log x \rceil \quad (x \geq 2). \quad (4.4)$$

Starting with the identifier bound $h = M$, apply the one-round reduction while $h > H_0$, replacing the common bound by $f(h)$. If $M = 1$ or already $M \leq H_0$, no such round is necessary. In the former case, distinct identifiers imply that there is at most one vertex. In every case the current proper labels can be viewed as elements of $[H_0]$ once the loop has ended.

For completeness, we prove that the loop takes at most $2 \log^* M + 2$ rounds, with constants independent of d . Set

$$T = (4A)^4.$$

For $x \geq 2^T$, write $y = \log x \geq T$. As $\lceil y \rceil \leq 2y$ for $y \geq 1$, monotonicity gives

$$f(f(x)) \leq A(\log(2Ay) + 1) \leq y = \log x. \quad (4.5)$$

Here is an explicit verification of the second inequality. For $y \geq T$, the function

$$g(y) = y - A(\log(2Ay) + 1)$$

has derivative $g'(y) = 1 - A/(y \ln 2) > 0$. Moreover,

$$\begin{aligned} A(\log(2AT) + 1) &= A(\log(2A) + 4 \log(4A) + 1) \\ &\leq 18A^2 + A < 256A^4 = T, \end{aligned}$$

where $\log z \leq z$ for $z \geq 1$, and $A \geq 32$. Thus $g(T) > 0$, proving (4.5).

As long as a pair of rounds starts with bound at least 2^T , either the loop stops during that pair, or its new bound is at most the logarithm of the bound at the start of the pair. More explicitly, let x_j be the bound after $2j$ executed rounds. If the first $j + 1$ pairs are executed and each starts with bound at least 2^T , then $x_{j+1} \leq \log x_j$; induction gives $x_j \leq \log^{(j)} M$, where $\log^{(j)}$ denotes j successive logarithms. Thus after at most $\log^* M$ pairs the loop has stopped or its bound has dropped below 2^T ; otherwise its bound would eventually be at most 1.

Once the bound is below 2^T , at most two more rounds are needed. Indeed, by monotonicity the first new bound is at most $f(2^T) = AT$, and the second is at most

$$\begin{aligned} f(AT) &\leq A(\log(AT) + 1) \\ &\leq 6A \log(4A) < 16A \lceil \log(4A) \rceil = H_0. \end{aligned}$$

The middle inequality follows from $\log(AT) = \log A + 4\log(4A) \leq 5\log(4A)$ and $\log(4A) \geq 1$. The loop stops earlier if it reaches H_0 at an intermediate round. This proves the claimed bound.

4.3 A final polynomial reduction to a quadratic palette

Choose q to be the smallest power of two at least $8u$, so

$$8u \leq q < 16u. \quad (4.6)$$

We use the polynomial graph construction of Erdős, Frankl, and Füredi [2, Example 3.2], as in Linial's color reduction [5, Lemma 4.2].

We will need a field $\mathbb{F}$ of order q . We recall its existence explicitly. Write $q = 2^r$. Over the two-element field, form a splitting field of $X^q - X$. Such a field can be constructed by repeatedly adjoining a root of an irreducible factor of degree at least two until all factors are linear: the quotient of a polynomial ring over a field by an irreducible polynomial is a field, because the Euclidean algorithm supplies an inverse modulo that polynomial for every nonzero residue class. At each step a further factor becomes linear, so a finite succession of such extensions suffices.

The derivative of $X^q - X$ in characteristic two is -1 , so in its splitting field the polynomial has exactly q distinct roots. These roots form a field: they contain 0 and 1, and are closed under addition and multiplication because repeated squaring gives $(a + b)^q = a^q + b^q$ and $(ab)^q = a^q b^q$. Additive inverses cause no problem in characteristic two. For a nonzero root a , the identity $(a^{-1})^q = (a^q)^{-1} = a^{-1}$ gives closure under inverses. Thus their set is a field with exactly q elements.

For a uniform common representation, enumerate pairs of binary operation tables on the ordered set $[q]$, with prescribed distinct labels for 0 and 1, and choose the first pair satisfying the field axioms. This finite search terminates by the existence argument. Field elements are ordered by their labels; this is merely an order for deterministic choices, not an order compatible with field arithmetic.

There are q^3 polynomials of degree at most two over this field, indexed by their triples of coefficients. This is enough to assign a distinct polynomial P_i to every temporary color $i \in [H_0]$. To check this, for integer $u \geq 2$ the inequality $u \leq 2^{u-1}$ follows by induction, and hence

$$\lceil \log(32u^2) \rceil \leq 2u + 3 \leq 4u.$$

Since $A = 8u^2$, we obtain

$$H_0 = 128u^2 \lceil \log(32u^2) \rceil \leq 512u^3 \leq q^3. \quad (4.7)$$

Use coefficient lexicographic order to make the assignment common and deterministic.

For each $i \in [H_0]$, put

$$H_i = \{(a, P_i(a)) : a \in \mathbb{F}\} \subseteq \mathbb{F}^2.$$

Each H_i has q elements. Distinct H_i, H_j intersect in at most two points, because $P_i - P_j$ is a nonzero polynomial of degree at most two. The root bound used here follows by division by $X - a$: a root a gives a factor $X - a$, and induction on degree bounds the number of distinct roots by the degree.

At most d outgoing neighbors can therefore cover at most $2d < q$ points of a vertex's own set H_i . In one more round, a vertex of temporary color i chooses the least pair in its H_i outside the sets corresponding to its outgoing neighbors. This is precisely the transformation

in (4.3) with the H_i in place of the S_i , so its correctness proof applies verbatim. Order $\mathbb{F}^2$ lexicographically by the common field labels and relabel its elements by $[q^2]$. The result is a proper ordered coloring with

$$K = q^2 < 256u^2 = 256(d+1)^2.$$

The previous loop took at most $2 \log^* M + 2$ rounds, and this last reduction takes one. This proves Proposition 4.1.

5 Proof of the main theorem

Run Proposition 4.1, then apply the algorithm of Proposition 2.2 using its output ranks. The list stage preserves membership in the actual input lists and separates every edge, by Lemma 3.4. Its composition with preprocessing takes at most

$$(2 \log^* M + 3) + (2K + 4) < 2 \log^* M + 512(d+1)^2 + 7 = O(\log^* M + d^2)$$

rounds. Under a polynomial identifier bound with fixed exponent, $\log^* M = O(\log^* n + 1)$, giving the stated bound in n .

All searches in preprocessing have finite domains, explicitly testable conditions, and proved existence of a successful object. All other choices are least elements of proved nonempty finite sets, with local port order resolving the distinguished-neighbor choice. The common parameters determine every phase length. These facts give one terminating deterministic program for all values of d and M .

Every communication step reads only outgoing-neighbor states. The algorithm uses no incoming-degree bound and no acyclicity assumption. In the list stage it compares and transmits only finitely many input names and their subsets; it never enumerates a global color universe. Longer representations can increase message size and local computation, which the stated model does not charge as rounds. This completes the proof of Theorem 2.1. $\square$

References

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